

Analytical Geometry

1. Coordinate geometry

- Cartesian coordinates (x, y) to polar coordinates (r, θ) :

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) \quad \text{in quadrant I}$$

$$= \pi - \tan^{-1}\left(\left|\frac{y}{x}\right|\right) \quad \text{in quadrant II}$$

$$= \pi + \tan^{-1}\left(\left|\frac{y}{x}\right|\right) \quad \text{in quadrant III}$$

$$= -\tan^{-1}\left(\left|\frac{y}{x}\right|\right) \quad \text{in quadrant IV}$$

- Polar coordinates (r, θ) to Cartesian coordinates (x, y) :

$$x = r \cos \theta$$

$$y = r \sin \theta$$

- Distance between two points in Cartesian coordinates $P(x_1, y_1)$ and $Q(x_2, y_2)$:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- Distance between two points in polar coordinates $P(r_1, \theta_1)$ and $Q(r_2, \theta_2)$:

$$d = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_1 - \theta_2)}$$

- Section formula (internal division):

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \quad y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

- Section formula (external division):

$$x = \frac{m_1 x_2 - m_2 x_1}{m_1 - m_2}, \quad y = \frac{m_1 y_2 - m_2 y_1}{m_1 - m_2}$$

- Midpoint formula:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- Area of triangle ABC with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$\Delta ABC = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

- Centroid G of triangle ABC with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

- Incenter I of triangle ABC with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ and side lengths a, b, c :

$$I = \left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c} \right)$$

- Excenter E_A opposite to vertex A:

$$E_A = \left(\frac{-ax_1 + bx_2 + cx_3}{-a + b + c}, \frac{-ay_1 + by_2 + cy_3}{-a + b + c} \right)$$

- Circumcenter O of triangle ABC:

$$O = \left(\frac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C} \right)$$

- Orthocenter H of triangle ABC:

$$H = \left(\frac{x_1 \tan A + x_2 \tan B + x_3 \tan C}{\tan A + \tan B + \tan C}, \frac{y_1 \tan A + y_2 \tan B + y_3 \tan C}{\tan A + \tan B + \tan C} \right)$$

2. Straight lines

- General form of a straight line:

$$ax + by + c = 0$$

- Slope (m) of a line passing through $(x_1, y_1), (x_2, y_2)$, making an angle θ with the positive x-axis:

$$m = \tan \theta = \frac{y_2 - y_1}{x_2 - x_1}$$

- Slope-intercept form (slope m , y-intercept c):

$$y = mx + c$$

- Point-slope form (passes through (x_1, y_1) with slope m):

$$y - y_1 = m(x - x_1)$$

- Two-point form (passes through (x_1, y_1) and (x_2, y_2)):

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

- Intercept form (x-intercept a , y-intercept b):

$$\frac{x}{a} + \frac{y}{b} = 1$$

- Normal (perpendicular) form (perpendicular distance from origin p , angle α with positive x-axis):

$$x \cos \alpha + y \sin \alpha = p$$

- Parametric / symmetric form (distance r from (x_1, y_1) along angle θ):

$$\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r$$

- Conditions for $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ to be coincident:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

- Conditions for $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ to be parallel to each other:

$$m_1 = m_2 \Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

- Equation of a line parallel to $ax + by + c = 0$:

$$ax + by + k = 0$$

- Conditions for $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ to be perpendicular to each other:

$$m_1m_2 = -1 \Rightarrow a_1a_2 + b_1b_2 = 0$$

- Equation of a line orthogonal to $ax + by + c = 0$:

$$bx - ay + k = 0$$

- Angle between two lines with slopes m_1 and m_2 :

$$\tan \theta = \pm \frac{m_1 - m_2}{1 + m_1m_2}$$

- Perpendicular distance from (x_1, y_1) to $ax + by + c = 0$:

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

- Perpendicular distance between two parallel lines $ax + by + c_1 = 0$, $ax + by + c_2 = 0$:

$$d = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$$

- Foot of perpendicular (h, k) from (x_1, y_1) to $ax + by + c = 0$:

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = -\frac{ax_1 + by_1 + c}{a^2 + b^2}$$

- Image / reflection (h, k) of (x_1, y_1) in line $ax + by + c = 0$:

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = -2\frac{ax_1 + by_1 + c}{a^2 + b^2}$$

- Equation of a line that bisects the angle between two intersecting lines $(a_1x + b_1y + c_1 = 0, a_2x + b_2y + c_2 = 0)$:

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

1. If $a_1a_2 + b_1b_2 > 0$:

- The + sign gives the obtuse angle bisector.

- The - sign gives the acute angle bisector.
- 2. If $a_1a_2 + b_1b_2 < 0$:
 - The + sign gives the acute angle bisector.
 - The - sign gives the obtuse angle bisector.
- Family of lines passing through the intersection of $L_1 = 0$ and $L_2 = 0$:

$$L_1 + \lambda L_2 = 0$$

- Conditions for concurrency of three lines ($a_1x + b_1y + c_1 = 0, a_2x + b_2y + c_2 = 0, a_3x + b_3y + c_3 = 0$):

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

- Angle between a pair of straight lines passing through the origin ($ax^2 + 2hxy + by^2 = 0$):

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{|a + b|}$$

3. Conics

3.1. Terms Related to Conics

- Focus (Plural: Foci): A fixed point used to define the conic section. The set of all points on the curve maintains a constant ratio regarding their distance to the focus relative to the directrix.
- Axis of Symmetry (Major/Transverse Axis): The line passing through the focus (or foci) and perpendicular to the Directrix, dividing the conic into two symmetrical halves.
- Directrix (Plural: Directrices): A fixed line perpendicular to the axis of symmetry.
- Latus Rectum: The chord that passes through a focus, runs perpendicular to the major/transverse axis, and has both endpoints lying on the curve.

- Vertex (Plural: Vertices): The point or points where the conic section intersects its principal axis of symmetry.
- Center: The midpoint of the segment connecting the two foci (applies to circles, ellipses, and hyperbolas; parabolas do not have a center).
- Eccentricity (e): A measure of how much the conic section deviates from being a circle.

$$e = \frac{\text{distance to focus}}{\text{distance to directrix}}$$

- Circle: $e = 0$
- Ellipse: $0 < e < 1$
- Parabola: $e = 1$
- Hyperbola: $e > 1$

3.2. General equation of a conic

- $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$
 If $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$, then it represents a pair of straight lines
 If $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} \neq 0$,
 $\Rightarrow h^2 - ab < 0, \quad a = b, h = 0 \rightarrow \text{Circle}$
 $\Rightarrow h^2 - ab < 0 \rightarrow \text{Ellipse}$
 $\Rightarrow h^2 - ab = 0 \rightarrow \text{Parabola}$
 $\Rightarrow h^2 - ab > 0 \rightarrow \text{Hyperbola}$
- General Expression of Conic (S): S represents the expression of the conic section equation when all terms are moved to one side so that the equation equals to zero. **e.g.** For a circle:

$$S = x^2 + y^2 + 2gx + 2fy + c$$

- Evaluating at a specific point (S_1): S_1 is evaluated by substituting the coordinates of a specific given value (x_1, y_1) directly into the expression S . **e.g.** For a circle:

$$S = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$$

- Tangent transformation operator (T): T is created by replacing the variables (x, y) in the expression S using a special set of rules with the give point (x_1, y_1):

Term in S	Term in T
x^2	xx_1
y^2	yy_1
$2x$	$x + x_1$
$2y$	$y + y_1$
$2xy$	$xy_1 + x_1y$
c	c

e.g. For a circle:

$$S = xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c$$

3.3. Universal Conic Properties (Applied to Any Conic $S = 0$)

- Position of a point (x_1, y_1) relative to conic S :
 - Outside: $S_1 > 0$ (for ellipse/circle)
 - On the curve: $S_1 = 0$
 - Inside: $S_1 < 0$ (for ellipse/circle)
- Equation of tangent at (x_1, y_1) on $S = 0$:

$$T = 0$$

- Chord of contact from an external point (x_1, y_1):

$$T = 0$$

- Pair of tangents from an external point (x_1, y_1):

$$S \cdot S_1 = T^2$$

- Chord bisected at a given point (x_1, y_1) :

$$T = S_1$$

- Polar of a point (x_1, y_1) with respect to $S = 0$:

$$T = 0$$

3.4. Circle

- General Equation: $x^2 + y^2 + 2gx + 2fy + c = 0$

- Center: $(-g, -f)$

- Radius: $r = \sqrt{g^2 + f^2 - c}$

- x-intercept length: $2\sqrt{g^2 - c}$

- y-intercept length: $2\sqrt{f^2 - c}$

- Touches x-axis: $g^2 = c$

- Touches y-axis: $f^2 = c$

- Touches both axes: $g^2 = f^2 = c$

- Center on x-axis: $f = 0$

- Center on y-axis: $g = 0$

- Center at origin: $g = 0, f = 0$

- Passes through origin: $c = 0$

- Touches x-axis at origin: $g = 0, c = 0$

- Touches y-axis at origin: $f = 0, c = 0$

- Central form:

$$(x - h)^2 + (y - k)^2 = r^2 \quad \text{where center} = (h, k), \text{radius} = r$$

- Parametric form:

- For $x^2 + y^2 = r^2$: $x = r \cos \theta, y = r \sin \theta \quad (0 \leq \theta < 2\pi)$

- For $(x - h)^2 + (y - k)^2 = r^2$: $x = h + r \cos \theta, y = k + r \sin \theta$

- Diameter form (endpoints (x_1, y_1) and (x_2, y_2)):

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

- Condition for tangency ($y = mx + c$ to $x^2 + y^2 = r^2$):

$$c = \pm r\sqrt{1 + m^2}$$

- Tangent in slope form: $y = mx \pm r\sqrt{1 + m^2}$

- Point of contact:

$$\left(\mp \frac{rm}{\sqrt{1+m^2}}, \pm \frac{r}{\sqrt{1+m^2}} \right)$$

- Equation of normal at point (x_1, y_1) :
 - For $x^2 + y^2 = r^2$: $yx_1 - xy_1 = 0$
 - For $x^2 + y^2 + 2gx + 2fy + c = 0$: $\frac{x-x_1}{x_1+g} = \frac{y-y_1}{y_1+f}$
- Length of tangent from external point (x_1, y_1) :

$$L = \sqrt{S_1}$$

- Angle θ between tangents from external point (x_1, y_1) :

$$\tan\left(\frac{\theta}{2}\right) = \frac{r}{\sqrt{S_1}}$$

- Area of triangle formed by tangents and chord of contact:

$$\Delta = \frac{r \cdot S_1^{\frac{3}{2}}}{S_1 + r^2}$$

- Director circle (locus of perpendicular tangents):
 - For $x^2 + y^2 = r^2$: $x^2 + y^2 = 2r^2$
- Pole of line $lx + my + n = 0$ with respect to $x^2 + y^2 = r^2$:

$$\left(-\frac{r^2 l}{n}, -\frac{r^2 m}{n} \right)$$

- Family of Circles:
 - Through intersections of $C_1 = 0$ and $C_2 = 0$: $C_1 + kC_2 = 0$ ($k \neq -1$)
 - Through intersections of circle $C = 0$ and line $L = 0$: $C + kL = 0$
 - Co-axial family: $S + \lambda L = 0$ or $S_1 + \lambda(S_1 - S_2) = 0$
- Radical axis / common chord of two circles $s_1 = 0$ and $s_2 = 0$:

$$S_1 - S_2 = 0$$

- Radical center of three circles $S_1 = 0, S_2 = 0, S_3 = 0$:
Intersection of $S_1 - S_2 = 0$ and $S_2 - S_3 = 0$.
- Angle of intersection between circles C_1 and C_2 :

$$\cos \theta = \frac{r_1^2 + r_2^2 - d^2}{2r_1 r_2} = \frac{2(g_1 g_2 + f_1 f_2) - (c_1 + c_2)}{2\sqrt{g_1^2 + f_1^2 - c_1}\sqrt{g_2^2 + f_2^2 - c_2}}$$

- Orthogonal intersection ($\theta = 90^\circ$): $2g_1 g_2 + 2f_1 f_2 = c_1 + c_2$
- Relative position of two circles (center distance $d = C_1 C_2$):
 - One inside another: $d < |r_1 - r_2|$ (0 common tangents)
 - Touch internally: $d = |r_1 - r_2|$ (1 common tangent)
 - Intersect at two points: $|r_1 - r_2| < d < r_1 + r_2$ (2 common tangents)
 - Touch externally: $d = r_1 + r_2$ (3 common tangents)
 - Separate: $d > r_1 + r_2$ (4 common tangents)
- lengths of common tangents:
 - Direct common tangent: $L_d = \sqrt{d^2 - (r_1 - r_2)^2}$
 - Transverse common tangent: $L_t = \sqrt{d^2 - (r_1 + r_2)^2}$
- Centers of similitude (divide center line in ratio $r_1 : r_2$):
 - Internal:

$$\left(\frac{r_1 x_2 + r_2 x_1}{r_1 + r_2}, \frac{r_1 y_2 + r_2 y_1}{r_1 + r_2} \right)$$

- External:

$$\left(\frac{r_1 x_2 - r_2 x_1}{r_1 - r_2}, \frac{r_1 y_2 - r_2 y_1}{r_1 - r_2} \right)$$

- Circle through 3 non-collinear points:

$$\begin{vmatrix} x^2 + y^2 & x & y & 1 \\ x_1^2 + y_1^2 & x_1 & y_1 & 1 \\ x_2^2 + y_2^2 & x_2 & y_2 & 1 \\ x_3^2 + y_3^2 & x_3 & y_3 & 1 \end{vmatrix} = 0$$

3.5. Parabola

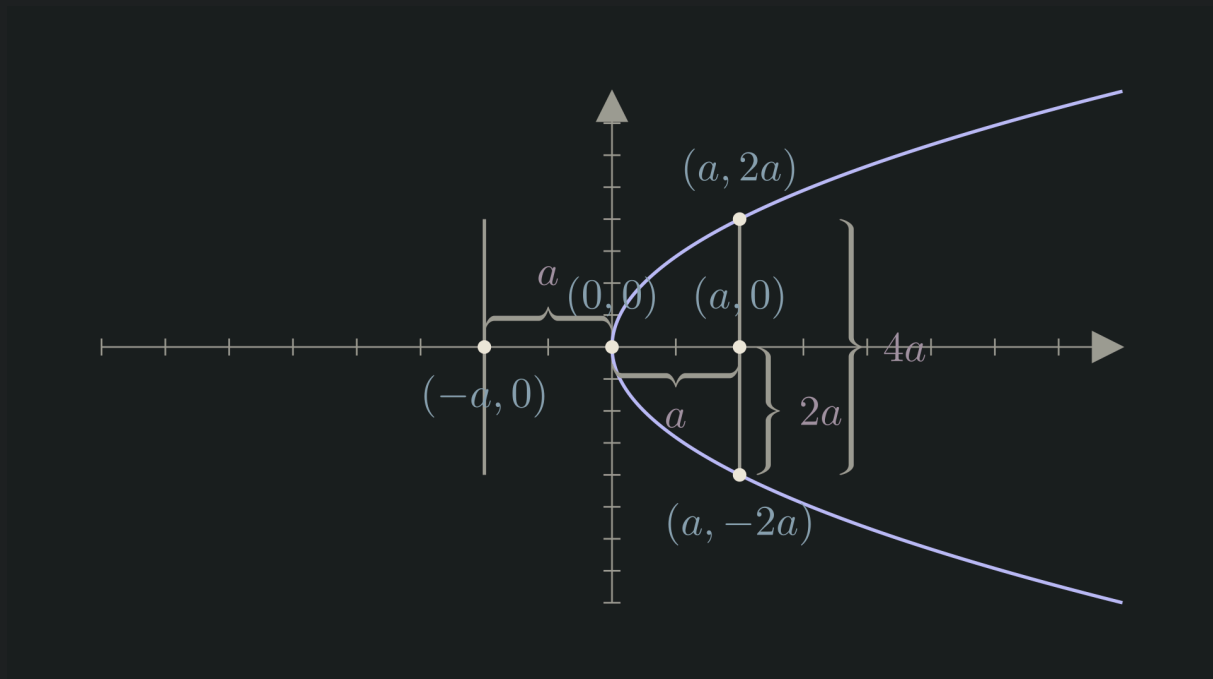


Figure 1: $y^2 = 4ax$

Description	$y^2 = 4ax$	$x^2 = 4ay$
Coordinates of vertex	$(0,0)$	$(0,0)$
Coordinates of focus	$(a,0)$	$(0,a)$
Axis-directrix intersection	$(-a,0)$	$(0,-a)$
Endpoints of latus rectum	$(a, \pm 2a)$	$(\pm 2a, a)$
Focal parameter (Focus to directrix)	$ 2a $	$ 2a $
Focal distance of point (x_1, y_1)	$x_1 + a$	$y_1 + a$
Length of latus rectum	$ 4a $	$ 4a $
Equation of directrix	$x = -a$	$y = -a$
Equation of axis	$y = 0$	$x = 0$
Equation of latus rectum	$x = a$	$y = a$
Equation of tangent at vertex	$x = 0$	$y = 0$

Equation of tangent at (x_1, y_1)	$yy_1 = 2a(x + x_1)$	$xx_1 = 2a(y + y_1)$
Equation of normal line at (x_1, y_1)	$y - y_1 = -\frac{y_1}{2a}(x - x_1)$ $y = -tx + 2at + at^3$	$x - x_1 = -\frac{x_1}{2a}(y - y_1)$ $x = -ty + 2at + at^3$
Condition for tangency $(y = mx + c)$	$c = \frac{a}{m}$	$c = -am^2$
Parametric equation	$x = at^2, y = 2at$	$x = 2at, y = at^2$
Eccentricity	$e = 1$	$e = 1$
Director Circle (Perpendicular Tangents Locus)	$x = -a$ (Directrix)	$y = -a$ (Directrix)

3.6. Ellipse

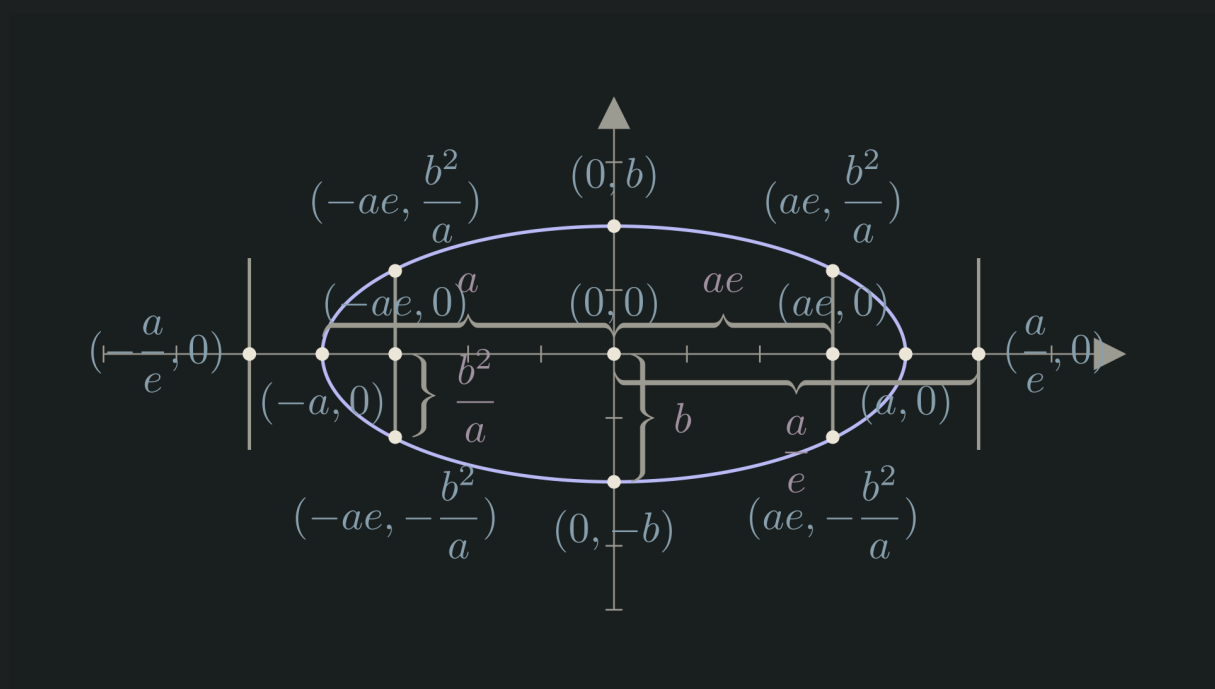


Figure 2: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$

Description	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, b > a$
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Coordinates of center	$(0, 0)$	$(0, 0)$
Coordinates of foci	$(\pm ae, 0)$	$(0, \pm be)$
Vertices / Major axis endpoints	$(\pm a, 0)$	$(0, \pm b)$
Minor axis endpoints	$(0, \pm b)$	$(\pm a, 0)$
Length of major axis	$2a$	$2b$
Length of minor axis	$2b$	$2a$
Length of latus rectum	$2\frac{b^2}{a}$	$2\frac{a^2}{b}$
Distance between foci	$2ae$	$2be$
Focal radii of point (x_1, y_1)	$a \pm ex_1$	$b \pm ey_1$
Distance between directrices	$2\frac{a}{e}$	$2\frac{b}{e}$
Equation of major axis	$y = 0$	$x = 0$
Equation of minor axis	$x = 0$	$y = 0$
Equation of directrices	$x = \pm \frac{a}{e}$	$y = \pm \frac{b}{e}$
Equation of latus rectum	$x = \pm ae$	$y = \pm be$
Equation of tangent at (x_1, y_1)	$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$	$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$
Equation of normal line at (x_1, y_1)	$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$	$\frac{b^2y}{y_1} - \frac{a^2x}{x_1} = b^2 - a^2$
Condition for tangency ($y = mx + c$)	$c = \pm \sqrt{a^2m^2 + b^2}$	$c = \pm \sqrt{b^2m^2 + a^2}$

Parametric equation	$x = a \cos \theta, y = b \sin \theta$	$x = a \cos \theta, y = b \sin \theta$
Eccentricity	$e = \sqrt{1 - \frac{b^2}{a^2}}$	$e = \sqrt{1 - \frac{a^2}{b^2}}$
Area	πab	πab
Director Circle	$x^2 + y^2 = a^2 + b^2$	$x^2 + y^2 = a^2 + b^2$

3.7. Hyperbola

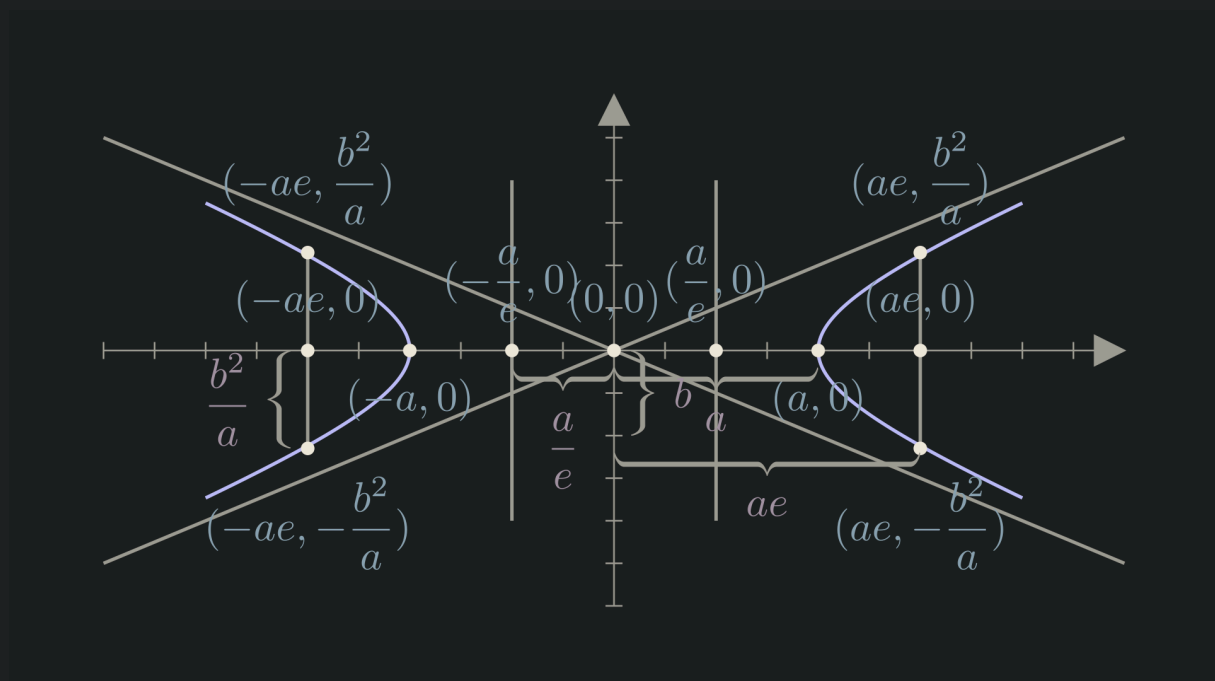


Figure 3: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Description	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$
Coordinates of center	(0, 0)	(0, 0)
Coordinates of foci	($\pm ae, 0$)	(0, $\pm be$)
Vertices / Transverse axis endpoints	($\pm a, 0$)	(0, $\pm b$)

Conjugate axis endpoints	$(0, \pm b)$	$(\pm a, 0)$
Length of transverse axis	$2a$	$2b$
Length of conjugate axis	$2b$	$2a$
Length of latus rectum	$2\frac{b^2}{a}$	$2\frac{a^2}{b}$
Distance between foci	$2ae$	$2be$
Focal radii of point (x_1, y_1)	$ ex_1 \pm a $	$ ey_1 \pm b $
Distance between directrices	$2\frac{a}{e}$	$2\frac{b}{e}$
Equation of transverse axis	$y = 0$	$x = 0$
Equation of conjugate axis	$x = 0$	$y = 0$
Equation of directrices	$x = \pm \frac{a}{e}$	$y = \pm \frac{b}{e}$
Equation of latus rectum	$x = \pm ae$	$y = \pm be$
Equation of asymptotes	$y = \pm \frac{b}{a}x$	$x = \pm \frac{a}{b}y$
Equation of tangent at (x_1, y_1)	$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$	$\frac{yy_1}{b^2} - \frac{xx_1}{a^2} = 1$
Equation of normal line at (x_1, y_1)	$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$ $ax \cos \theta + by \cot \theta = a^2 + b^2$	$\frac{b^2y}{y_1} + \frac{a^2x}{x_1} = a^2 + b^2$ $by \cos \theta + ax \cot \theta = a^2 + b^2$
Condition for tangency ($y = mx + c$)	$c = \pm \sqrt{a^2m^2 - b^2}$	$c = \pm \sqrt{b^2 - a^2m^2}$
Parametric equation	$x = a \sec \theta, y = b \tan \theta$	$x = a \tan \theta, y = b \sec \theta$
Eccentricity	$e = \sqrt{1 + \frac{b^2}{a^2}}$	$e = \sqrt{1 + \frac{a^2}{b^2}}$

Director Circle	$x^2 + y^2 = a^2 - b^2 \quad (a > b)$	$x^2 + y^2 = b^2 - a^2 \quad (b > a)$
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• Rectangular Hyperbola

Characteristic	Standard Form $x^2 - y^2 = a^2$	Rotated Form $xy = c^2$
Relationship	Derived from standard form with $a = b$	Derived by rotating $x^2 - y^2 = a^2$ by 45° where $c^2 = \frac{a^2}{2}$
Axis orientation	Transverse axis along x-axis	Transverse axis along line $y = x$
Coordinates of vertices	$(\pm a, 0)$	$(\pm c, \pm c)$
Coordinates of foci	$(\pm a\sqrt{2}, 0)$	$(\pm c\sqrt{2}, \pm c\sqrt{2})$
Length of latus rectum	$2a$	$2\sqrt{2}c$
Equation of asymptotes	$y = \pm x$	$x = 0, y = 0$
Parametric form	$x = a \sec \theta, y = a \tan \theta$	$x = ct, y = \frac{c}{t}$
Eccentricity	$e = \sqrt{2}$	$e = \sqrt{2}$